

Task 4: Context-Aware Chatbot Using LangChain & RAG

Objective

Build a conversational chatbot that remembers context and retrieves external information using Retrieval-Augmented Generation (RAG). The chatbot answers questions based on a knowledge corpus and maintains conversational memory across multiple turns.

Files

File	Description
notebook.ipynb	Complete RAG pipeline with multi-turn conversation demo
src/corpus_loader.py	Knowledge corpus creation and text chunking
src/rag_pipeline.py	FAISS index, embeddings, and retrieval
src/chatbot.py	Conversational memory and RAG chatbot class
outputs/figures/	Generated plots (PNG, 200 DPI)

Quick Start

```
pip install -r ../requirements.txt
jupyter notebook notebook.ipynb
```

Note: The first run will download `all-MiniLM-L6-v2` (~80MB). This may take 2-3 minutes.

Key Results

- **Knowledge Corpus:** 15 AI/ML documents
- **Text Chunks:** ~40 chunks (80 words each, 15-word overlap)
- **Embedding Model:** all-MiniLM-L6-v2 (384-dim)
- **Vector Store:** FAISS IndexFlatL2
- **Retrieval Top-1 Accuracy:** ~75-88%
- **Retrieval Top-3 Accuracy:** ~88-100%
- **Conversational Memory:** 5-turn sliding window

Tech Stack

- Python 3.10+
- sentence-transformers (Hugging Face)
- FAISS (Facebook AI Similarity Search)
- pandas, numpy
- matplotlib, seaborn
- Jupyter Notebook